NISHANTH VEDURUVADA

AI Platform Architect

Agentic Platforms • AI Governance • Enterprise AI Modernization • Zero Trust Architecture

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PROFESSIONAL SUMMARY

AI Platform Architect with 11+ years across software engineering, cloud and solution architecture, and application security — a path from building production mobile and data-intensive systems, through enterprise security architecture, into the platform and governance layer that lets autonomous systems run inside regulated enterprises. Currently AI Engineer Advisor at NTT DATA, leading architecture and technical governance for 8–10 enterprise AI modernization programs across the Syntphony product ecosystem, directing a team of AI engineers and data scientists. Previously sole architect of the AI Trust Fabric — a NIST 800-207 Zero Trust and AI governance architecture for a national-scale insurance data ecosystem under insurance-regulator oversight, registered as an industrial design with the Government of India. Designs agentic systems as enterprise platforms rather than prototypes: agent topology and tool protocols (MCP, A2A) over identity, authorization, audit and observability, on cloud infrastructure built to carry them — in insurance, telecommunications and energy.

SELECTED ACHIEVEMENTS

- **Reference architecture for a reusable agentic platform.** Standardized MCP as the tool-integration primitive and A2A for cross-domain agent requests, with domain agents composed as plugins on a shared platform — so identity, audit, rate limiting and human approval are inherited as platform behaviour instead of rebuilt per engagement.
- **Registered industrial design, Government of India.** Sole architect of the AI Trust Fabric — NIST 800-207 Zero Trust extended with continuous adaptive trust scoring and an AI agent governance layer, for a national-scale insurance data ecosystem under regulator oversight. Co-registered as Design No. 485611-001.
- **Peer-reviewed publication on AI agent security.** Co-author, *MCP Guardian* — middleware adding authentication, per-token rate limiting and WAF-style tool-call scanning to the Model Context Protocol without modifying individual MCP servers, evaluated at roughly 3–4 ms median latency overhead (10–15%).
- **Zero Trust migration across 5,000+ users.** Led network segmentation and a SASE rollout replacing perimeter-based access with identity-centric controls — the security foundation the later AI governance architecture is built on.

CORE COMPETENCIES

Enterprise & Solution Architecture · AI Platform Architecture · Agentic & Multi-Agent Systems · AI Governance · Zero Trust Architecture · Cloud Architecture · Technical Leadership

PROFESSIONAL EXPERIENCE

NTT DATA
Dec 2025 – Present

AI Engineer Advisor

Architecture and technical governance for AI modernization across the Syntphony product ecosystem — Syntphony AI, Open Telecom Networks, Insurance Cloud Migration, Stations, and Commerce & Payments.

- **Agentic Platform Architecture:** Engagement teams were each rebuilding tool integration, agent messaging, audit and rate limiting — duplicated effort, inconsistent security posture. Defined a reusable reference architecture standardizing MCP as the tool primitive and A2A for cross-domain requests, with domain agents (OSS assurance, inventory, fulfilment, RAN automation) composed as plugins, so a new domain ships as a plugin rather than a new stack.
- **Agent Identity & Trust Boundaries:** Established per-agent OAuth2 workload identity in place of borrowed user credentials, action-scope enforcement restricting each agent to its declared capabilities, per-agent and per-tool rate limits, full tool-call audit logging, and a human-in-the-loop approval gateway for high-risk actions — with OpenTelemetry tracing per agent turn and tool call, so agent behaviour is reconstructable after the fact.
- **Enterprise Retrieval Architecture:** Designed a dual-path knowledge assistant separating live analytics from indexed content — semantic-model queries executed as the calling user with row-level security passthrough, alongside permission-scoped document retrieval, routed by a query classifier. Metric values are never embedded, so figures cannot go stale; the same multi-tenant pattern scopes access per business unit and country on a client-facing deployment.
- **Multi-Agent Operational Intelligence:** Architected cross-domain root-cause analysis over five siloed OSS domains (RAN, Core, Transport, Inventory, Assurance). A query-planning agent decomposes natural-language questions and dispatches to domain-specialist agents holding their own API tools, with a correlation layer synthesizing answers cited to source, element, metric and timestamp — specialist agents over one monolith, because each domain carries its own data model and metric vocabulary.
- **Legacy Modernization:** Architected an MCP-based code intelligence system over mainframe COBOL and JCL estates for insurance cloud migration, built on a structure-aware parser and a static dependency graph. The model queries the live codebase through MCP tools rather than an embedded index — the estate changes as modernization proceeds, and answers must cite exact program, paragraph and line. Enforced a truth boundary so parsed structure remains the system of record and model output stays advisory against it, with a context-assembly layer that minimizes what reaches the model to hold token cost down as the estate grows.
- **Architecture Governance & Technical Leadership:** Lead architecture reviews, decision records and trade-off analysis for a team of AI engineers, senior engineers and data scientists; own enterprise AI architecture standards, technical strategy, stakeholder alignment and mentoring.

Levo.ai — **API security platform** Jan 2025 – Dec 2025

AI Enterprise Architect

Client: national-scale insurance data ecosystem (India)

Sole architect of the AI Trust Fabric for a national-scale insurance data ecosystem aggregating records from across the country's insurance industry, under insurance-regulator oversight and data-localization requirements.

- **API Security Platform:** Before the client placement, worked on the platform's AI capabilities — including an MCP server exposing client API telemetry as queryable insights. Production MCP work at a security vendor, alongside co-authoring peer-reviewed research on securing that same protocol.
- **Zero Trust Architecture:** Designed the full NIST 800-207 structure — policy administration, decision, enforcement and information points as separate concerns — with enforcement deployed as an API gateway intercept, so every call, human or agent, traverses one trust boundary rather than relying on per-application checks.
- **Adaptive Trust Engine:** Replaced binary allow/deny with continuous trust scoring across identity confidence, device posture, behavioural baseline, data classification and context, driving allow / step-up-authentication / deny decisions with per-data-class thresholds, cached in second-scale windows to trade freshness against latency on the hot path.
- **AI Agent Identity & Governance:** Gave agents their own workload identity rather than borrowed user credentials, with action-scope enforcement, per-agent rate limiting, prompt-injection detection, full decision audit, and human escalation above a risk threshold, with trust signals for whether a session's tool-call pattern matches the agent's declared purpose.
- **Offline AI Capability:** Architected fully offline inference with Ollama, and a natural-language policy intelligence layer over insurance regulation, to satisfy no-internet-egress environments — a hard constraint that would otherwise have blocked the entire governance layer.
- **Architecture Conformance Validation:** Encoded the organisation's regulatory obligations — India's Digital Personal Data Protection Act among them — as a structured rule taxonomy, then assessed deployed servers against those rules and against the Zero Trust architecture as designed. Findings such as out-of-support .NET runtimes raised Jira tickets automatically against the owning department, so drift became tracked work rather than a report. Locally hosted models throughout, so regulated personal data never left the environment being assessed.
- **Explainability & Audit:** Wired traceable rationale into every access decision — which policy matched, which trust signals fired — against an immutable audit log, making AI-assisted decisions defensible under regulatory audit.

Arcstream Technologies
Aug 2022 – Jan 2025

Technical Security Lead

- **Enterprise Security Architecture:** Led Zero Trust network segmentation and a SASE migration covering 5,000+ users, moving the trust boundary from the network perimeter to identity.
- **Secure Delivery & Governance:** Hardened CI/CD pipelines and established architecture governance across engineering teams, embedding security review into the delivery path.
- **AI Agent Security Research:** Co-authored the peer-reviewed *MCP Guardian* under this affiliation — where the security work turned toward AI systems.

Reezanorp Technologies
Dec 2016 – Jul 2022

Senior Software Engineer

- **Applied ML & Data Engineering:** Built machine learning, analytics, computer vision, ETL and enterprise automation systems in Python — the move from application development into data-intensive systems, and the engineering foundation the later architecture work is built on.

Corpus Media Labs
Mar 2015 – Dec 2016

Software Engineer

PUBLICATIONS & INTELLECTUAL PROPERTY

MCP Guardian: A Security-First Layer for Safeguarding MCP-Based AI Systems
Computer Science & Information Technology (CS&IT), Vol.15 No.9, pp.107–121, 2025 · DOI 10.5121/csit.2025.150908 · Co-author

AI Trust Fabric — Secure Insurance Data Management Device
Registered Design No. 485611-001, Government of India · Co-registrant

OPEN SOURCE

agent-governance-patterns
github.com/NishVed/agent-governance-patterns · Apache-2.0
Reference implementations of enterprise agent governance controls: a disposition ladder resolving each agent output from confidence, actor authority, action risk and segregation-of-duties, and an exception-over-silence result type. Framework-agnostic, no dependencies, tested.

TECHNICAL SKILLS

AI & Agentic Architecture: Agentic AI, Multi-Agent Systems, Model Context Protocol (MCP), Agent-to-Agent (A2A), Tool Calling, Agent Identity, Action-Scope Enforcement, Human-in-the-Loop, Agent Observability, Prompt-Injection Defence, Explainability

AI Platforms & LLMOps: Large Language Models (LLM), Generative AI (GenAI), Retrieval-Augmented Generation (RAG), Semantic Search, Vector Databases, Azure AI Search, Prompt Engineering, LLMOps, Azure OpenAI, Azure AI Foundry, Ollama

Frameworks: LangGraph, CrewAI, LangChain, Semantic Kernel, AutoGen, FastMCP

Security & Governance: Zero Trust Architecture, NIST 800-207, Policy Decision and Enforcement Points (PAP/PDP/PEP/PIP), AI Governance, Identity and Access Management (IAM), Microsoft Entra ID, OAuth 2.0, OpenID Connect (OIDC), Zero Trust Network Access (ZTNA), Secure Access Service Edge (SASE), API Security, Data Loss Prevention (DLP), Row-Level Security (RLS)

Cloud & Platform: Microsoft Azure, Azure API Management, Azure Functions, Azure Logic Apps, Kubernetes (AKS), Docker, Microservices, OpenTelemetry, CI/CD, DevSecOps

Distributed Systems & Data: Event-Driven Architecture, Apache Kafka, Azure Service Bus, Cosmos DB, Microsoft Fabric, Power BI, Microsoft Graph, SQL, ETL, Machine Learning

Engineering & Leadership: Python, FastAPI, REST APIs, Legacy Modernization, COBOL/JCL Analysis, Solution Architecture, Enterprise Architecture, Technical Leadership, Mentoring, Stakeholder Management

EDUCATION

Bachelor of Technology (B.Tech), Electronics & Communication Engineering — Jawaharlal Nehru Technological University, India